From Songlines to the Cloud

Episode 2: Measuring the World

[Cold open]

About twenty-two centuries ago, in the third century BCE, the most important library in the ancient world was run by a man named Eratosthenes of Cyrene. As far as we know, he's the person who gave us the word *geography*. And he did something with it that's still one of the best demonstrations of spatial thinking ever devised.

In the version handed down by a later writer, Cleomedes, it goes like this. He knew that at noon on the summer solstice, in the town of Syene, which is modern Aswan, the sun stood directly overhead. At the same moment in Alexandria, further north, an upright stick cast a short shadow. Measure the angle of that shadow and it comes to one-fiftieth of a full circle. So, he reasoned, the distance from Alexandria to Syene must be one-fiftieth of the way around the Earth.

He took that distance as five thousand stadia. Fifty times five thousand: two hundred and fifty thousand stadia for the whole planet. Then he nudged it up to two hundred and fifty-two thousand, apparently because that number divides cleanly by every whole number from one to ten. Seven hundred stadia to a degree. Here was a man who liked a tidy spreadsheet.

[Beat]

How close did he get? That depends entirely on how long his stadion was, and nobody knows for sure. If he meant a shorter, Egyptian stadion, his answer comes out under one percent off the real figure. If he meant the longer Attic stadion, he was about sixteen percent too big. But here's the thing. Whatever the unit, the *method* was sound: two places, one moment, a difference in the sun's angle, a known distance between them. That's geodesy. Two thousand years of measuring the Earth start right there, with a stick and a shadow.

[Theme]

This is *From Songlines to the Cloud*. I'm Paul. Last time, we saw spatial knowledge move out of people's heads and into the archives of the state. This time, it gets measured. We'll go from Eratosthenes to the end of the eighteenth century, when, for the first time, the whole Earth could be placed on a single grid. And we'll find that the story runs through Alexandria, Xi'an, Sicily and Florence before it gets anywhere near London.

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Greek geography begins in Miletus, on the coast of what's now Turkey. Ancient sources say that Anaximander, in the sixth century BCE, was the first Greek who dared to draw the inhabited world on a map. He pictured the Earth as a drum, or a column, floating unsupported in space. Hecataeus, a generation or two later, improved the map and wrote a book called the *Circuit of the Earth*; Eratosthenes himself considered him the first geographer. And Herodotus, our friend from last episode, couldn't resist mocking the neat, perfectly circular ocean on his predecessors' maps.

After Eratosthenes, the bar went up. Hipparchus insisted that places should be fixed by astronomy, by latitude and longitude, not by travelers' guesses, and he even proposed using lunar eclipses to work out differences in longitude. Strabo, around the turn of the millennium, went the other way and wrote seventeen books of practical, human-centered description.

And then, in Alexandria again, around 150 CE, came Claudius Ptolemy. His *Geography* is one of the most influential books ever written about the Earth, and if you work with spatial data, it will feel oddly familiar. Ptolemy gave coordinates for roughly eight thousand places: latitude measured from the equator, longitude from a prime meridian out in the Fortunate Isles, off the coast of Africa. He gave instructions for drawing a world map and twenty-six regional maps. And he offered three different map projections for flattening the sphere onto a plane.

None of Ptolemy's own maps survive. Every "Ptolemy map" you've ever seen is a reconstruction, drawn later from his coordinates; the oldest we have are Byzantine, from the late thirteenth century. In other words, what came down to us wasn't a map at all. It was a dataset, with a rendering spec.

[Beat]

It was also a dataset with bugs. Ptolemy adopted a measurement of the Earth that was too small, smaller than Eratosthenes', and he stretched Eurasia across about half the globe. Hold on to those two errors. Thirteen centuries later, they're going to get a Genoese sailor very excited.

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It's tempting to tell this story as a European relay race, with the baton passing from Greece to Rome to the Renaissance. It isn't one. Measured mapping grew up independently in more than one place, and the ideas traveled between cultures.

In China, a scholar named Pei Xiu, who lived in the third century CE and whom the historian Joseph Needham called the "father of scientific cartography in China," set out six principles of mapmaking. They include proportional scale, orientation, measured road distances, and adjustments for heights and for winding routes. It's a recognizably modern checklist. His eighteen-sheet regional maps are lost.

But in Xi'an there's a stone stele carved in 1136, the *Yu Ji Tu*, the "Map of the Tracks of Yu." China is laid out on a precise grid, each square standing for a hundred *li*, about fifty kilometers. Put it next to a modern map and the coastline and the great river systems, the Yellow River and the Yangtze, are immediately recognizable. Analysis shows the east–west lines are accurate while the north–south lines drift from true meridians, which is what happens when you lay a flat grid over a round Earth. Specialists warn against romanticizing it as modern coordinate geodesy. Still, carved in stone, centuries before anything comparable in Europe, it's a remarkable object.

Then there's the Islamic world, which did far more than keep Ptolemy safe until Europe was ready for him. In the ninth century, al-Khwārizmī, the mathematician whose name gives us the word *algorithm*, produced a revised Ptolemaic gazetteer, *The Book of the Picture of the Earth*. He cut Ptolemy's badly overstretched Mediterranean down by about ten degrees; a later astronomer, al-Zarqālī, got it to the correct figure. And al-Khwārizmī showed the Atlantic and Indian Oceans as open seas, not landlocked basins. Around the same time, the Caliph al-Ma'mūn sent surveyors into the Syrian desert to measure a degree of the meridian and work out the size of the Earth again from scratch.

About two centuries later, al-Bīrūnī found a cleverer way. He climbed a mountain in the Punjab, measured how far the horizon dipped below the level, and used a bit of trigonometry to calculate the radius of the Earth. His result, as best we can reconstruct his units, was within about six-tenths of one percent of the modern value. Nobody in the West matched that precision until the sixteenth century.

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The most celebrated map of the medieval Islamic world was commissioned by a Christian king.

In 1138, Roger the Second, the Norman king of Sicily, brought the scholar al-Idrīsī to his court. For about fifteen years, al-Idrīsī interviewed travelers, and here's the part any researcher will appreciate: he cross-checked their accounts and kept what agreed. He combined that with the classical sources. In January 1154 he finished what's known as the *Book of Roger*. It organized the world into Ptolemy's seven climate zones, cut into ten sections each, giving seventy sectional maps, plus a circular map of the world. That world map was engraved for Roger on a disc of silver about two meters across.

It had south at the top. It was the most accurate map of the world in its day, and it stayed in use for three centuries.

The same culture produced geography on foot. Ibn Baṭṭūṭa spent some twenty-nine years traveling across Africa, Asia and Europe, and dictated his account, the *Riḥla*, in 1355: travel writing as a way of gathering geographic knowledge.

[Optional: an aside on Ibn Baṭṭūṭa's name turning up here in Dubai, if you'd like to use it.]

South Asia gives us a useful corrective. The dominant Indian tradition was cosmographic: Mount Meru at the center of the world, surrounded by concentric rings of continents and oceans. These world models appear in diagrams, in temple art, and on cosmographic globes, rather than in survey maps. That isn't an absence of geography. It's a reminder that representing space has always served sacred purposes as well as practical ones.

Medieval Europe, meanwhile, ran two traditions at once. One was symbolic. The great *mappae mundi* were encyclopedias and theology in map form, not navigation tools. The Hereford Mappa Mundi, from around 1300, is the

largest one that survives: a single sheet of vellum about a meter and a half tall, with something like five hundred drawings, including about four hundred and twenty towns, biblical scenes, beasts, plants, and peoples. East is at the top, and Jerusalem sits in the middle.

The other tradition appeared, in one historian's words, "rather suddenly" in the late thirteenth century: portolan charts. They show the coastlines of the Mediterranean and the Black Sea with astonishing accuracy, criss-crossed with networks of compass lines. They were made by and for sailors and merchants.

[Beat]

The two coexisted because they answered different questions. One asked where humanity stands in a divinely ordered universe. The other asked how to make port before the weather turned.

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Around 1400, a Byzantine scholar named Manuel Chrysoloras brought the Greek text of Ptolemy's *Geography* to Florence. By 1406 or 1407 there was a Latin translation, and once printing arrived it spread fast: the first printed edition with maps came out in Bologna in 1477, and dozens more editions followed.

What Ptolemy gave Renaissance Europe wasn't really a correct picture of the world. It was a toolkit: the idea of the coordinate grid, the graticule, and the idea of the projection. That's the conceptual kit of modern mapping.

It also came with those two bugs. A small Earth, and an Asia stretched far to the east. Put them together and the ocean between Europe and Asia looks a lot narrower than it is. Christopher Columbus probably leaned on exactly that picture when he argued a westward passage to Asia was feasible. It may be the most productive error in the history of geography.

Exploration turned maps into strategic assets. Spain and Portugal kept secret master charts and revised them constantly as new coastlines came in. In 1507 Martin Waldseemüller published a world map that was the first to use the name "America," after Amerigo Vespucci. The only surviving copy is in the Library of Congress.

In 1569, Gerardus Mercator published a wall map on eighteen sheets with a title that told you exactly who it was for: a new and enlarged description of the Earth, "corrected for the use of sailors." His projection spaces the parallels further and further apart toward the poles, so that a line of constant compass bearing comes out perfectly straight. If you're steering a ship, that's gold. Mercator never published how he built it; an Englishman, Edward Wright, worked out the tables in 1599, and it became the standard nautical projection.

And the price, of course, is that high latitudes balloon. Every argument you've ever had with a class about why Greenland looks the size of Africa starts here, in a map that was never meant to be hung on a classroom wall.

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Then states took over measurement itself. The Cassini survey of France was the first national topographic survey built on triangulation, a framework of precisely measured triangles laid across the whole country. Louis the Fifteenth ordered it in 1747. Four generations of the Cassini family worked on it, producing around a hundred and eighty sheets over the next half-century and more. When the Revolution came, the government seized the plates for the war office in 1793. The map was a military asset now.

There's a much-repeated story, probably too good to be strictly true, that when the earlier, more accurate coastal surveys shrank France on the map, Louis the Fourteenth complained that the Academy's surveyors had cost him more territory than his enemies ever had.

That left one great problem: longitude. Latitude is relatively easy; you can read it from the sun and the stars. Longitude is about time. To know how far east or west you are, you need to compare the time where you are with the time at some reference point, at the same instant. On a rolling ship, with eighteenth-century clocks, that was close to impossible, and ships were being lost.

After a run of maritime disasters, including the wreck of a fleet off the Scilly Isles in 1707, the British Parliament passed the Longitude Act of 1714. It offered twenty thousand pounds for a method accurate to within half a degree on a voyage across the Atlantic.

The man who cracked it was John Harrison, a carpenter who became a clockmaker. He built three large sea clocks, and then, in 1759, he finished something completely different: a watch, H4, small enough to hold in your hand. On a trial voyage to Jamaica starting in November 1761, it lost about five seconds in eighty-one days. On a second trial to Barbados in 1764, it was off by thirty-nine seconds over forty-seven days: three times better than the prize required.

[Beat]

The Board of Longitude paid him half. He had to get the rest, in 1773, with help from King George the Third and Parliament. Harrison is one of history's great examples of a person being right well before the committee was ready to admit it.

A copy of his watch, made by Larcum Kendall, sailed with Captain James Cook on his second and third voyages into the Pacific. Cook called it "our trusty friend the Watch." Across his voyages, using readings of the moon against the stars, rigorous coastal surveys and, from the second voyage on, the watch, Cook fixed New Zealand, eastern Australia, and Hawai'i onto the world grid with unprecedented accuracy.

Think about where that puts us. The same ocean that Pacific navigators had crossed for millennia by reading the swells was now being pinned to a coordinate system built in Europe.

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Here's the hinge of the whole series. By about 1800, the Earth was measurable, navigable, and mappable in one coordinate system. Projection, triangulation, the chronometer. Notice who paid for each of them: sailors, empires and tax collectors, just as on the Nile. The stakes drove the tools.

Everything that comes next sits on this foundation. Academic geography is built on it. So, a century and a half later, is GIS. When your phone puts a blue dot on a map, it's standing on Eratosthenes' shadow and Harrison's watch.

Next time, geography goes to university. Isaac Newton edits a geography textbook. Immanuel Kant teaches the subject forty-nine times. A Prussian decree calls for a chair of geography at every Prussian university. And the new discipline picks up some baggage it will spend a century trying to put down.

I'm Paul. This has been *From Songlines to the Cloud*. Thanks for listening.

[Theme out]